# The Finite-Harmonic Formal Second-Order Tangent Cone of Compact Weyl–Maxwell Gravity

Moment Maps, Exponential-Polynomial Corrections, and Bounded Resonances

GPT-5.6.sol* Claude Fable 5†

Working draft, 19 July 2026

**Abstract**

Linearized solutions of a nonlinear gauge theory need not be tangent to exact solutions. We classify this failure at second order for finite harmonic configurations of Weyl–Maxwell gravity on the compact magnetically supported product $\mathbb{R}_t \times S^1 \times S^2$. The certified linear inventory contains the ordinary Einstein–Maxwell branches, two additional fourth-order branches in each generic parity, exceptional dipoles, a homogeneous generalized block, and axial twist modes.

For spatially periodic corrections with finite exponential-polynomial time dependence, the complete second-order obstruction space is five-dimensional. Its covectors are precisely the Taub pairings with time translation, circle translation, and the three sphere rotations. Thus the complete finite-support exponential-polynomial tangent cone is

$$\mathcal{Z}_2^{\mathrm{ep}} = \{u : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}.$$

This is a necessary-and-sufficient theorem in the declared correction class, not merely a list of examples.

Bounded corrections obey additional conditions. Polynomial-growth source coefficients and nonzero-shell adjoint pairings survive independently of the five moment maps. We classify the complete standard generalized-zero subcone, prove that electric-charge and flat-Wilson oscillator columns are bounded-removable, and determine the full circumference column: a nonzero circle-radius tangent is compatible with a finite oscillator precisely when that oscillator has zero circle momentum. A repaired full-time circumference-velocity fixture also exposes a nonzero polar polynomial obstruction that was invisible at $t = 0$; its full $a, d$-cross ideal on the $\ell = 2, k = 0$ extra block is now classified. On the complete global-plus-$\ell = 2, k = 0$-extra carrier, the bounded cone then collapses exactly to the standard static-global cone: no nonzero extra amplitude survives. At one tuned nonzero momentum, by contrast, we classify the complete axisymmetric all-primary bounded cone. Its only shell resonance cuts out two mixed axial–polar Einstein-minus sheets, and the lower-frequency extra primary widens the exact interval on which positive-primary occupations can balance their energy and momentum. The remaining oscillator columns are stated as open gates. A successor theorem promotes the standard-branch part of this calculation to every $\ell \geq 2$: the exact two-parity resonance matrix has two mixed sheets and two one-sided planes, and charge balance reduces the complete tuned axisymmetric bounded cone to the origin together with the two mixed sheets on an explicit occupation interval. A separate two-absolute-momentum computation has certified all 108 axisymmetric $L = 4$ branch-basis coefficients and all 56 declared nonaxisymmetric $L = 1, 3$ coefficients; every individual basis

*OpenAI model and principal programme author.

†Anthropic model and computational coauthor. The project was commissioned and coordinated by Asger Alstrup Palm (asger@area9.dk), who initiated the questions, exercised editorial control, and serves as corresponding human contact, but claims no technical authorship.

fixture is obstructed, while amplitude-level cancellations remain open. Consequently formal exponential-polynomial extendibility is a moment-map theorem, whereas bounded quasiperiodic extendibility is a strict resonance problem.

All results are LOCAL-ALGEBRAIC and REDUCED-MODE. They do not establish an all-orders family, a causal retarded correction, a final residual quotient, or a quantum state space.

## AI authorship and computational accountability

The named AI systems developed the programme, derivations, symbolic implementations, independent verification architecture, claim audits, and this manuscript. Asger Alstrup Palm commissioned and coordinated the work and is the corresponding human contact. Every theorem stated as certified below is bound to an exact machine-readable certificate and a separately implemented verifier. Exact rational or algebraic arithmetic is used for all ranks, cokernels, polynomial reductions, and source remainders. The manuscript remains conditional on final human review of its mathematical scope, references, and exposition.

## Contents

# 1 Question, answer, and claim boundary

Let $\Phi(s)$ be a putative curve of solutions of the Weyl–Maxwell equations $\mathcal{E}(\Phi) = 0$, expanded as

$$\Phi(s) = \bar{\Phi} + su + s^2 v + O(s^3).$$

The first two equations are

$$\mathcal{L}u = 0, \qquad \mathcal{L}v = -\mathcal{E}^{(2)}_{\bar{\Phi}}(u,u), \qquad \mathcal{L} := D\mathcal{E}_{\bar{\Phi}}, \qquad \mathcal{E}^{(2)}_{\bar{\Phi}} := \frac{1}{2}D^2\mathcal{E}_{\bar{\Phi}}. \tag{1}$$

The first equation describes the linear solution space. The second asks whether the quadratic source lies in the image of the linearized operator in a specified function class. Changing that function class changes the answer: a resonant source may have a finite exponential-polynomial secular primitive but no bounded quasiperiodic primitive.

The paper answers the following finite-harmonic question.

> For the complete certified linear inventory on the compact product, which real finite-support tangents admit a spatially periodic second-order correction, and which additional conditions appear when that correction is required to remain bounded in time?

The answer has two layers:

$$u \in \mathcal{Z}_2^{\mathrm{ep}} \iff \mu_H = \mu_{P_x} = \mu_{J_1} = \mu_{J_2} = \mu_{J_3} = 0, \tag{2}$$

$$u \in \mathcal{Z}_2^{\mathrm{qp}} \iff u \in \mathcal{Z}_2^{\mathrm{ep}}, \quad \mathcal{P}_{j,r}(u) = 0, \quad \mathcal{R}_{j,a}(u) = 0. \tag{3}$$

Here $\mathcal{P}_{j,r}$ are positive polynomial-growth coefficients and $\mathcal{R}_{j,a}$ are reduced adjoint pairings on nonzero characteristic shells. Both obstruction ledgers are complete in the declared finite carrier. The paper gives several complete columns of the second line but does not claim a classification of the common quadratic zero locus of all bounded obstructions.

## 1.1 Dependency typing

The spacetime is Lorentzian, but the proof is an exact harmonic and finite-module calculation. It carries the dependency tags

LOCAL-ALGEBRAIC REDUCED-MODE.

No theorem below carries LORENTZIAN-CAUSAL. In particular, a finite exponential-polynomial right inverse is not a retarded Green operator, and a bounded quasiperiodic correction is not an all-orders nonlinear solution.

# 2 Relation to classical linearization stability

The structural origin of the five quadratic conditions is classical. Brill and Deser exhibited nonintegrable linearized solutions on closed spatial sections [1]; Fischer and Marsden formulated linearization stability as a nonlinear functional-analytic problem [2]; and Moncrief identified spacetime symmetries with the adjoint-kernel constraints governing second-order integrability [3, 4]. Arms treated the coupled Einstein–Maxwell problem on the appropriate electromagnetic bundle [5]. The Arms–Marsden–Moncrief momentum-map normal form then described the local solution space near a symmetric point as a smooth factor times the zero set of a homogeneous quadratic map [6, 8]. Higher-curvature Taub constraints and their second-order interpretation have also been developed in general form [10, 11].

Accordingly, this paper does *not* claim to discover the general principle "tangent cone equals moment-map zero." Its contribution is an explicit exhaustive realization in the declared fourth-order Weyl–Maxwell model: the complete certified finite carrier, a proof that exactly five covectors exhaust the exponential-polynomial cokernel, and a bounded refinement by independent polynomial-growth and characteristic-shell obstructions. The latter belongs to the broader problem of secular perturbation theory [12], but the concrete $\{\mu, \mathcal{P}, \mathcal{R}\}$ ledger and exceptional mixed resonance are model-specific results.

The relation is conceptual rather than a claim that the classical functional-analytic normal-form theorem has been reproved for this theory. Those theorems use Sobolev-completed initial-data manifolds, a constraint map with suitable splitting/Fredholm properties, gauge slices, and control of the full infinite-dimensional adjoint kernel. Here we instead prove an algebraic surjectivity and cokernel theorem on a complete *declared finite harmonic module* of the fourth-order Weyl–Maxwell equations. We do not establish the hypotheses needed to promote it to an infinite-mode Sobolev normal form. Conversely, the classical structural theorem does not by itself supply our block inventory, five-covector sufficiency statement, or bounded $\mathcal{P}/\mathcal{R}$ refinement.

# 3 Theory and common background

We use the compactified Plebański–Hacyan product

$$M = \mathbb{R}_t \times S^1_L \times S^2, \tag{4}$$

with unit sphere curvature and a parallel unit magnetic flux on a fixed compact $U(1)$ bundle. With the conventions of the companion linear phase-space paper, the background solves both Einstein–Maxwell and Weyl–Maxwell theory. The latter has local gauge group

$$\mathrm{Diff}(M) \times \mathrm{Weyl}(M) \times U(1).$$

The spatial slice $\Sigma = S^1 \times S^2$ is compact. Its background stabilizer algebra contains

$$H = \partial_t, \qquad P_x = \partial_x, \qquad J_1, J_2, J_3 \in \mathfrak{so}(3).$$

For a linear tangent $u$, the corresponding covariant phase-space moment maps are quadratic forms

$$\mu_X(u) = \frac{1}{2}\Omega(u, \mathcal{L}_X u), \qquad X \in \{H, P_x, J_1, J_2, J_3\}.$$

On a closed slice, the covariant Taub bridge identifies these with the pairings of the quadratic field-equation source against the stabilizer cokernel covectors.

**Proposition 3.1** (Taub moment maps on ordinary and generalized modes)**.** *The five displayed automorphisms lift to the fixed magnetic bundle and act symplectically on every certified linear block. For every linear solution, including the homogeneous and twist Jordan blocks,*

$$\left\langle \zeta_X, \mathcal{E}^{(2)}_{\bar\Phi}(u,u) \right\rangle_\Sigma = \mu_X(u) = \frac{1}{2}\Omega_\Sigma(u, \mathcal{L}_X u), \qquad X \in \{H, P_x, J_1, J_2, J_3\}. \tag{5}$$

*The right side is independent of the closed Cauchy slice $\Sigma$. There is no sixth $U(1)$ Taub covector in the fixed-bundle problem.*

*Proof.* Twice differentiating the action Noether identity pairs the quadratic constraint source with the adjoint reducibility parameter and gives (5). Because $u$ and $\mathcal{L}_X u$ solve the linearized equations, their Lee–Wald current is conserved. The difference between its integrals on two closed Cauchy slices is therefore the integral of an exact spacetime divergence and vanishes. This argument uses neither a frequency eigenbasis nor boundedness, so it applies to polynomial Jordan solutions.

Concretely, time translation acts on the homogeneous coordinates by

$$(a, b, c, d, Q_e, W_x) \longmapsto (b, 0, d, 2a, 0, Q_e)$$

and on each twist pair by $(A, B) \mapsto (B, 0)$. Substitution into the conserved Cauchy form gives, after omission of the common positive harmonic normalizations,

$$\mu_H^{\ell=0} = -a^2 - b^2 + bd - Q_e^2, \tag{6}$$

$$\mu_H^{\text{twist}} = 2|B|^2, \qquad \mu_{\mathbf{J}}^{\text{twist}} = -4A \times B. \tag{7}$$

Although the representatives contain powers of $t$, these expressions are constant; the apparent time-dependent contributions cancel in the conserved current.

The rotations lift as bundle automorphisms $\widehat{J}_a = (J_a, \chi_a)$, with $\iota_{J_a}\bar{F} + \mathrm{d}\chi_a = 0$ patchwise. A constant $U(1)$ parameter acts trivially on connection differences and is a reducibility parameter, not an independent Hamiltonian stabilizer. Finally, fixing the Chern class of the magnetic bundle forbids a harmonic magnetic variation. These facts leave exactly the five covectors in (5). □

# 4 The complete finite harmonic carrier

The real finite-support carrier $V_{\text{fin}}$ is the direct sum of the following certified linear blocks:

1. generic Einstein and extra primaries for every $\ell \geq 2$, parity, magnetic label, and compact momentum;
2. standard and extra exceptional dipoles for every compact momentum;
3. the axial twist Jordan block;
4. the six-coordinate homogeneous generalized block; and
5. the electric and Wilson global directions on the fixed bundle.

Reality pairs opposite frequencies and momenta. A finite input therefore produces a finite quadratic source. Its complete output decomposition is indexed by

$$(L, M, K, \Omega, \text{parity}).$$

Angular Clebsch–Gordan selection gives finitely many $L$, while temporal and circle products give finite sums of input frequencies and momenta.

**Definition 4.1** (Finite correction modules)**.** Let $\mathscr{X}_{\text{ep}}$ be the spatially periodic fields whose time coefficients are finite sums of $t^p e^{-i\Omega t}$, with $p \in \mathbb{N}_0$ and a finite set of real frequencies. Let $\mathscr{X}_{\text{qp}} \subset \mathscr{X}_{\text{ep}}$ be the finite bounded quasiperiodic subclass with $p = 0$. Define

$$\mathcal{Z}_2^{\mathscr{X}} = \{u \in V_{\text{fin}} : \exists v \in \mathscr{X},\ \mathcal{L}v = -\mathcal{E}^{(2)}(u, u)\}.$$

Every element of $\mathscr{X}_{\text{ep}}$ is smooth, but $\mathscr{X}_{\text{ep}}$ is not the full $C^\infty$, Sobolev, distributional, or retarded function space. The subscripts “ep” and “qp” are therefore load-bearing. A secular coefficient is finite at every finite time but prevents a uniformly bounded perturbative expansion. It may signal a nearby family with shifted frequency, a required renormalization of parameters, or genuine resonant growth. The certificates decide only these two finite-module dispositions.

# 5 Finite exponential-polynomial tangent-cone theorem

The sufficiency statement rests on a simple algebraic fact which we state explicitly because it is the step that distinguishes $\mathscr{X}_{\mathrm{ep}}$ from $\mathscr{X}_{\mathrm{qp}}$.

**Lemma 5.1** (Exponential-polynomial surjectivity)**.** *Let $P(D_t) \neq 0$ be a constant-coefficient scalar differential operator. For every polynomial $f(t)$ and real $\Omega$, there is a polynomial $g(t)$ such that*

$$P(D_t)\big(g(t)e^{-i\Omega t}\big) = f(t)e^{-i\Omega t}. \tag{8}$$

*If $-i\Omega$ is a root of $P$ of multiplicity $m$, one may take $\deg g \leq \deg f + m$. Consequently every nonzero invariant factor of a matrix operator over the fibre ring $K[D_t]$ is surjective on $\mathscr{X}_{\mathrm{ep}}$; only zero Smith factors contribute to its $\mathscr{X}_{\mathrm{ep}}$-cokernel.*

*Proof.* Conjugation by $e^{-i\Omega t}$ replaces $P(D_t)$ by $P(D_t - i\Omega)$. Factor

$$P(z - i\Omega) = z^m Q(z), \qquad Q(0) \neq 0.$$

The operator $D_t^m$ is surjective on polynomials, increasing the required degree by at most $m$. On each finite-dimensional polynomial-degree filtration, $Q(D_t) = Q(0)(1 + N)$ with $N$ nilpotent, hence has the finite inverse $Q(0)^{-1} \sum_r (-N)^r$. For a matrix operator, unimodular Smith transformations preserve $\mathscr{X}_{\mathrm{ep}}$ and reduce the assertion to its scalar invariant factors. □

The complete output ledger after Noether restriction and local Diff $\times$Weyl $\times\, U(1)$ reduction is as follows. Here $p$ and $q$ are respectively the extra and Einstein characteristic polynomials; "zero factors" means persistent factors in the $\mathscr{X}_{\mathrm{ep}}$ cokernel, not roots of a nonzero characteristic polynomial.

| Output stratum | Reduced operator or invariant factors | Zero factors | Cokernel interpretation |
|---|---|---|---|
| Generic $L \geq 2$, nonstatic | $1, 1, p, pq$; off-shell inverse and Lemma 5.1 on $p/q$ shells | 0 | none |
| Generic $L \geq 2$, static | $p = -(3\Lambda + 3\kappa^2 - 2)/3 < 0$ and $q > 0$ for $\Lambda \geq 6$ | 0 | none |
| Exceptional $L = 1$, nonstatic | invertible off shell; nonzero factors on $\Omega^2 - K^2 = 4, 4/3$ | 0 | none in $\mathscr{X}_{\mathrm{ep}}$ |
| Scalar $L = 0$, $(\Omega, K) \neq (0, 0)$ | exact modulo local gauge and Noether identities | 0 | none |
| Homogeneous $L = 0$, $(\Omega, K) = (0, 0)$ | constraints plus $D_t^4(C - K)$ and $D_t^2 A_x$ | 2 | $\zeta_H, \zeta_{P_x}$ |
| Axial dipole $L = 1$, $(\Omega, K) = (0, 0)$ | twist polynomial operator; polar block invertible | 3 | $\zeta_{J_1}, \zeta_{J_2}, \zeta_{J_3}$ |
| Remaining certified global/twist outputs | contained in the preceding $L = 0, 1$ blocks | 0 additional | none additional |

For completeness, put $\kappa := K_{\mathrm{out}}$ for the output circle momentum (unrelated to the homogeneous function $K(t)$ below). The static generic entries use

$$p = -\frac{3\Lambda + 3\kappa^2 - 2}{3} < 0, \qquad q = \Lambda^2 + 2\Lambda\kappa^2 - 2\Lambda + \kappa^4.$$

Putting $s = \Lambda - 6 \geq 0$ and $u = \kappa^2 \geq 0$ gives

$$q = s^2 + 2su + u^2 + 10s + 12u + 24 > 0,$$

so no static $L \geq 2$ zero factor is hidden in the table.

For the homogeneous source, quadratic products have degree at most six. The two dynamical invariants therefore have the explicit polynomial right inverses

$$D_t^4\left(\frac{t^{r+4}}{(r+1)(r+2)(r+3)(r+4)}\right) = t^r, \qquad D_t^2\left(\frac{t^{r+2}}{(r+1)(r+2)}\right) = t^r, \quad 0 \leq r \leq 6.$$

This displays rather than merely asserts the zero-block surjectivity.

**Theorem 5.2** (Complete finite-support exponential-polynomial cone)**.** *For the complete carrier $V_{\text{fin}}$,*

$$\boxed{\mathcal{Z}_2^{\text{ep}} = \{u \in V_{\text{fin}} : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}.} \tag{9}$$

*Every point of this cone admits a real, spatially periodic second-order correction in $\mathscr{X}_{\text{ep}}$, and each of the five equations is necessary.*

*Proof.* Finite input support and Clebsch–Gordan selection produce finitely many output labels $j = (L, M, K, \Omega, \text{parity})$, with temporal polynomial degree at most six. The table exhausts those labels. Lemma 5.1 removes every nonzero invariant factor, including every nonzero-frequency resonance. The displayed polynomial primitives remove the homogeneous dynamical rows. Exactly five zero factors remain, so

$$\operatorname{coker} \mathcal{L}_{\text{ep}} = \operatorname{span}\{\zeta_H, \zeta_{P_x}, \zeta_{J_1}, \zeta_{J_2}, \zeta_{J_3}\}.$$

Pairing the second-order equation with them and applying Proposition 3.1 proves necessity. Conversely, when all five moment maps vanish, every output source lies in the blockwise image. The finite corrections and their conjugates assemble a real element of $\mathscr{X}_{\text{ep}}$, proving sufficiency. □

*Remark* 5.3. The theorem covers arbitrary finite sums across the certified generic, exceptional, homogeneous, and twist blocks. It is not a theorem for the full smooth, Sobolev, distributional, infinite-mode, or causal function space.

# 6 Complete bounded-obstruction decomposition

On $\mathscr{X}_{\text{qp}}$, differentiation cannot produce a positive power of $t$, and a resonant pure exponential has no bounded quasiperiodic primitive. For each finite output block $j$, let $\mathcal{P}_{j,r}$ extract the coefficient of $t^r e^{-i\Omega_j t}$ for $r > 0$. After all such coefficients vanish, let $\mathcal{R}_{j,a}$ pair the remaining pure exponential with an exact basis of the reduced left kernel of $\mathcal{L}_j(-i\Omega_j)$. This definition includes the dynamical reduced left kernel at $\Omega_j = 0$. The stabilizer covectors inside the genuine zero block are split off and recorded separately as $\mu_X$; no other zero-root covector is silently identified with them.

**Theorem 6.1** (Complete finite bounded-obstruction ledger)**.** *For every $u \in V_{\text{fin}}$, a second-order correction in $\mathscr{X}_{\text{qp}}$ exists if and only if*

$$\mu_H(u) = \mu_{P_x}(u) = \mu_{J_i}(u) = 0, \qquad \mathcal{P}_{j,r}(u) = 0, \qquad \mathcal{R}_{j,a}(u) = 0 \tag{10}$$

*for all finite output blocks and all corresponding indices. These are the complete linear obstruction types in the declared finite module. The common quadratic zero set of the resulting functionals is only partly classified.*

*Proof.* If a bounded quasiperiodic $v$ exists, then $\mathcal{L}v$ is also bounded and quasiperiodic, so every positive polynomial coefficient $\mathcal{P}_{j,r}$ vanishes. For each remaining pure-frequency block, finite-dimensional linear algebra says that $\mathcal{L}_j(-i\Omega_j)v_j = S_j$ is solvable precisely when $S_j$ annihilates its reduced left kernel. At the genuine zero block those pairings split into the five Taub moment maps and any remaining dynamical zero-root functionals $\mathcal{R}_{j,a}$; at nonzero shells they are the finite-frequency $\mathcal{R}_{j,a}$. Off-shell blocks are invertible. Conversely, these conditions put every finite block source in the corresponding image, and the finite collection of solutions and conjugates assembles an element of $\mathscr{X}_{\mathrm{qp}}$. ☐

The certificates contain a moment-map-zero fixture with a nonzero $\mathcal{P}_{j,2}$ and another with a nonzero $\mathcal{R}_{j,a}$. Thus both new families are independent of the five global charges, and $\mathcal{Z}_2^{\mathrm{qp}} \subsetneq \mathcal{Z}_2^{\mathrm{ep}}$.

For finite generic oscillatory inputs the complete bounded $K = \Omega = 0$ receiver can be made fully explicit. Its nonzero source map is

$$(\mu_H, \mu_{P_x}, \mu_{J_1}, \mu_{J_2}, \mu_{J_3}, R_c), \qquad R_c = \frac{1}{2}\sum_j k_j^2 h_j.$$

Here $h_j$ is the complete normalized Hermitian current block, including cross terms within a degenerate shell. The second formal homogeneous mean covector is the Wilson-acceleration row; its quadratic source vanishes by the integrated Maxwell identity. The polar $L = 1$ zero block contributes no physical cokernel, the axial block contributes exactly the three lifted rotations, and every static $L \geq 2$ block is reduced-invertible. Thus the six displayed conditions are necessary and sufficient for bounded inversion of the zero-frequency source. On the candidate-13 two-fibre carrier, joining them to the eighteen finite-frequency coefficients gives the complete formula

$$\mathcal{Z}_{2,13}^{\mathrm{qp}} = \{\mu_H = \mu_{P_x} = \mu_{J_1} = \mu_{J_2} = \mu_{J_3} = R_c = \mathcal{R}_{13,1} = \cdots = \mathcal{R}_{13,18} = 0\}.$$

**Theorem 6.2** (Candidate-13 bounded-origin theorem)**.** *On the complete real candidate-13 generic two-fibre carrier,*

$$\mathcal{Z}_{2,13}^{\mathrm{qp}} = \{0\}.$$

*The smooth exponential-polynomial cone does not collapse: it remains the common zero of the five stabilizer moment maps and contains the explicit three-occupation Einstein–extra fixture.*

*Proof.* Put

$$D = -\frac{8}{5L}\mu_H - \frac{3}{2L\sqrt{\rho}}\mu_{P_x} - \frac{2}{\rho}R_c, \qquad \rho = \frac{-250 + 461\sqrt{10}}{2132}.$$

In occupation variables this is

$$D = \frac{2}{5}\sum \omega^2 h - \frac{3}{8}\sum n\omega h - \sum n^2 h, \qquad n \in \{1, -2\}.$$

The Einstein-minus current is negative, while the extra and Einstein-plus current blocks are positive in both parities. Since $1/2 < \rho < 3/5$, exact rational bounds give positive coefficients at least $379/1000$ and $23/80$ on the two signed Einstein-minus fibres, and at least $101/250$ and $38/45$ on the corresponding extra fibres. The Einstein-plus frequencies are larger and their common positive-branch coefficient is increasing. Thus $D$ is a strictly positive sum of the absolute-current Gram occupations for every parity and $m$. Vanishing of $\mu_H, \mu_{P_x}, R_c$ forces every occupation, hence the tangent, to vanish. The rotations and eighteen resonance equations are then automatic. In the smooth correction class $R_c$ and the finite-frequency resonances have secular inverses, so only the five moment maps remain. ☐

The earlier three-occupation fixture has $R_c < 0$, consistently placing it in the smooth cone but outside the bounded cone. The theorem is a real positivity result on this exact circumference; it makes no identification with another collision fibre and does not classify the complex zero variety.

# 7 Complete standard generalized-zero subcone

The coordinates in the next theorem are concrete perturbations, not abstract labels. In coordinates $(t, x, \theta, \phi)$, write

$$h_{xx} = C(t), \qquad h_{\theta\theta} = K(t), \qquad h_{\phi\phi} = K(t)\sin^2\theta, \qquad a_x = W_x + Q_e t, \tag{11}$$

$$K(t) = a + bt, \qquad C(t) = at^2 + \frac{b}{3}t^3 + c + dt. \tag{12}$$

For an orthonormal real $\ell = 1$ basis $(Y_{1a}, X_a)$, the twist block is

$$h_{xA} = (A_a + B_a t)X_A^a, \qquad a_x = -(A_a + B_a t)Y_{1a}. \tag{13}$$

| Coordinate | Linear representative | Time behaviour | Geometric meaning |
|---|---|---|---|
| $a$ | sphere scale $K = a$ with induced $C = at^2$ | constant/quadratic | homogeneous radion/Jordan seed |
| $b$ | $K = bt$, $C = bt^3/3$ | linear/cubic | homogeneous Jordan partner |
| $c$ | $C = c$ | constant | circle circumference modulus |
| $d$ | $C = dt$ | linear | circumference velocity |
| $Q_e$ | $a_x = Q_e t$ | linear | uniform electric tangent |
| $W_x$ | $a_x = W_x$, $\mathrm{d}a = 0$ | constant | flat circle Wilson line |
| $A \in \mathbb{R}^3$ | twist $h_{xA}, a_x$ | constant | lifted $SO(3)$ mapping-torus holonomy |
| $B \in \mathbb{R}^3$ | same twist carrier | linear | twist velocity/Jordan partner |

**Theorem 7.1** (Standard generalized-zero bounded cone)**.** *On the complete standard generalized-zero carrier with homogeneous coordinates $(a, b, c, d, Q_e, W_x)$ and axial twist position/velocity $(A, B)$, bounded second-order extension holds exactly on*

$$a = b = Q_e = 0, \qquad B = 0, \qquad c, d, W_x \textit{ arbitrary}, \qquad A \in \mathbb{R}^3 \textit{ arbitrary}. \tag{14}$$

*The surviving source has an explicit bounded correction.*

The triangular positive-degree ideal supplies the proof. The complete homogeneous positive-degree coefficients are

| Equation row | coefficient of $t$ | coefficient of $t^2$ |
|---|---|---|
| $E_{11}$ | $15ab$ | $15b^2/2$ |
| sphere trace | $-15ab/2$ | $-15b^2/4$ |
| Maxwell $x$ row | $Q_e a$ | $Q_e b/2$ |

The remaining leading polynomial tensor is $\mathrm{STF}(B \otimes B)t^2$, whose squared real coefficient is $2|B|^4/3$. They force $b = B = 0$ and $Q_e a = 0$. Intersecting with the five moment maps leaves

$$\mu_H = -(a^2 + Q_e^2), \qquad \mu_{P_x} = 0, \qquad \mu_{\mathbf{J}} = 0,$$

and hence forces $a = Q_e = 0$ on this carrier. The remaining homogeneous source vanishes, while constant twist position is tangent to an exact lifted mapping-torus family and has a constant polar correction.

Explicitly, on the universal cover identify

$$(x, p) \sim (x + L, \exp(A^a J_a)p), \qquad p \in S^2, \tag{15}$$

and lift the rotation to the magnetic bundle with its patchwise Maxwell compensator. The local product fields descend by covariance. Differentiating this exact quotient family at $A = 0$ gives the constant twist representative; the stored quadratic correction is its constant polar $L = 2$ Taylor term.

**Corollary 7.2** (Universal polynomial elimination). *For every point of the complete finite-support bounded cone,*

$$b = 0, \qquad B = 0, \qquad Q_e a = 0.$$

*Oscillatory products cannot cancel these zero-frequency polynomial rows.*

# 8 Natural transport columns

Several apparently dangerous global–oscillator products are derivatives of exact families and can therefore be classified without solving each harmonic source independently.

**Proposition 8.1** (Electric and Wilson columns). *For every certified nonzero-frequency standard or extra primary, the electric-charge cross source has a bounded correction obtained by differentiating electromagnetic duality,*

$$F_\theta = \cos\theta F + \sin\theta \star_g F.$$

*At the background,*

$$h_{\text{cross}} = 0, \qquad f_{\text{cross}} = \star_{\bar{g}} f + (D_g \star)[h] F_{\text{bg}}.$$

*This is the mixed derivative at the purely magnetic background, not a claim that the finite duality orbit remains in one magnetic bundle component. Its sphere period vanishes for $\ell \geq 1$, so this mixed correction itself lifts globally as a fixed-bundle connection difference. Every Wilson–oscillator source vanishes because $\delta A = W_x dx$ has $\delta F = 0$ and the local action depends on the connection through $F$.*

This removes the complete $Q_e$ and $W_x$ oscillator columns from the bounded frontier. It does not remove the separate homogeneous condition $Q_e a = 0$.

**Theorem 8.2** (Complete circumference column). *Let $c$ vary the circle radius and let a certified oscillator have compact momentum $k$. The $c$-times-oscillator source admits a bounded correction if and only if*

$$c = 0 \qquad \text{or} \qquad k = 0.$$

*For $k \neq 0$, the radius family gives a correction in $\mathscr{X}_{\text{ep}}$ but none in $\mathscr{X}_{\text{qp}}$.*

Indeed, writing $R^2 = 1 + \eta c$,

$$\omega_R^2 = \frac{k^2}{R^2} + m_{\text{branch}}^2, \qquad \partial_\eta \omega_R\big|_0 = -\frac{ck^2}{2\omega}.$$

Differentiating the exact family produces

$$\partial_\eta e^{-i\omega_R t}\big|_0 = \frac{ick^2}{2\omega} t e^{-i\omega t}.$$

To make the shell inference explicit, write the reduced matrix equation as $L(\omega_R, R)u_R = 0$, choose a left on-shell vector $u^\dagger$, and differentiate at $R = 1$. The Feynman–Hellmann identity is

$$\left\langle u^\dagger, (\partial_\eta L)u \right\rangle = -\omega_0' \left\langle u^\dagger, (\partial_\omega L)u \right\rangle = \frac{ck^2}{2\omega} \left\langle u^\dagger, (\partial_\omega L)u \right\rangle. \tag{16}$$

The last factor is the current-normalized shell form. It is nonzero on each standard branch and nondegenerate on every extra multiplicity block. Hence the particular circumference source, not merely some source, has nonzero adjoint pairing whenever $ck \neq 0$ and the corresponding mode coefficient is nonzero. This is a genuine $\mathcal{R}_{j,a}$ obstruction, not a polynomial $\mathcal{P}_{j,r}$ obstruction. At $k = 0$, $\omega_0' = 0$ and differentiating the covariant circle indices gives an ordinary same-frequency correction.

# 9 Exceptional resonance as an independent obstruction

Moment-map cancellation does not guarantee bounded or even nonsecular second-order closure in a restricted exceptional carrier. For the axisymmetric exceptional $\ell = 1, k = 0$ axial and polar amplitudes $(a_x, a_p)$, a twist velocity can cancel the time-translation moment map. The two parity self-sources can also be tuned to cancel their even $L = 2$ resonance. But the required interior point has $a_x a_p \neq 0$, and the mixed axial $L = 2$ source has exact nonzero adjoint pairing

$$-\frac{8\sqrt{3}}{9}.$$

The boundary axes retain their individual bounded resonant self-obstructions. Thus the only axisymmetric two-polarization exceptional configuration in this carrier admitting a correction in $\mathscr{X}_{\mathrm{qp}}$ is $a_x = a_p = 0$. Nonzero configurations on the moment-map cone still admit the secular $\mathscr{X}_{\mathrm{ep}}$ corrections guaranteed by Theorem 5.2. The example proves that bounded resonant cokernel equations contain information independent of global charges; it does not contradict the exponential-polynomial theorem.

# 10 A complete tuned nonzero-momentum all-primary cone

The preceding exceptional result is a no-go on a restricted carrier. The next fixture shows that a bounded mixed cone can instead survive after every primary at the same angular and compact-momentum fibre is admitted. Fix

$$\ell = 2, \qquad m_A = 0, \qquad k^2 = 2\sqrt{3} - \frac{7}{6}, \tag{17}$$

choosing the circle circumference so that $\pm k$ are allowed compact momenta. The three positive input frequencies are

$$\omega_-^2 = \frac{29}{6}, \qquad \omega_e^2 = 2\sqrt{3} + \frac{25}{6}, \qquad \omega_+^2 = \frac{29}{6} + 4\sqrt{3}, \qquad \omega_- < \omega_e < \omega_+. \tag{18}$$

Here $q_-$ and $q_+$ denote the two Einstein primaries, while $X$ denotes the complete extra-primary multiplicity. Include both momentum signs, both axial and polar $q_-$ amplitudes, arbitrary normalized $X$ and $q_+$ multiplicities, and one constant twist position. No nonaxisymmetric input is included.

Write $A_\pm, P_\pm$ for the axial and polar $q_-$ coefficients at momenta $\pm k$. The exact 140-row collision census covers every output $L = 1, 2, 3, 4$, momenta $K = 0, \pm 2k$, and every zero, sum, and difference frequency of $q_-$, $X$, and $q_+$. Its unique characteristic hit is the polar extra shell

$$L = 4, \qquad K = 0, \qquad \Omega = 2\omega_-, \tag{19}$$

generated only by the $q_- q_-$ product. Its two independent adjoint pairings are nonzero multiples of

$$A_+ A_- - 3P_+ P_-, \qquad A_+ P_- - A_- P_+. \tag{20}$$

Their complete complex zero set consists of the two mixed sheets

$$A_\pm = \sigma\sqrt{3}\, P_\pm, \qquad \sigma \in \{+1, -1\}, \tag{21}$$

and the two one-sided planes $A_+ = P_+ = 0$ or $A_- = P_- = 0$.

On a mixed sheet define the nonnegative Einstein-minus occupations

$$N_\pm = -h_-(A_\pm e_A + P_\pm e_P, A_\pm e_A + P_\pm e_P), \tag{22}$$

and let $E_\pm$ and $B_\pm$ be the total nonnegative occupations in the extra and Einstein-plus multiplicities. The complete Hamiltonian and circle momentum equations are

$$\omega_e^2(E_+ + E_-) + \omega_+^2(B_+ + B_-) = \omega_-^2(N_+ + N_-), \tag{23}$$

$$\omega_e(E_+ - E_-) + \omega_+(B_+ - B_-) = \omega_-(N_+ - N_-). \tag{24}$$

**Theorem 10.1** (Tuned axisymmetric all-primary bounded cone)**.** *In the carrier just declared, a bounded finite-quasiperiodic second-order correction exists precisely at the origin or on one of the two mixed sheets* (21) *with*

$$\frac{1 - r_e}{1 + r_e} \le \frac{|P_+|^2}{|P_-|^2} \le \frac{1 + r_e}{1 - r_e}, \qquad r_e := \frac{\omega_-}{\omega_e}, \tag{25}$$

*and with nonnegative positive-primary occupations satisfying* (23)–(24)*. Their phases and multiplicity factorizations are otherwise arbitrary. The declared constant twist coordinate remains free and bounded-removable.*

*Proof.* Equations (20) give the four components displayed above. On either one-sided plane, positivity and $r_e < 1$ force the negative occupation to vanish when $\mu_H = \mu_{P_x} = 0$; hence only the origin survives. On a mixed sheet, the raw axial and polar weights cancel. At fixed positive-branch energy, the largest possible compact momentum is supplied by the lower positive frequency $\omega_e$, so (23)–(24) have a nonnegative solution if and only if

$$|N_+ - N_-| \le r_e(N_+ + N_-), \tag{26}$$

which is equivalent to (25). Sufficiency is explicit: set $B_+ = B_- = 0$ and choose

$$E_+ + E_- = r_e^2(N_+ + N_-), \qquad E_+ - E_- = r_e(N_+ - N_-). \tag{27}$$

The interval makes both $E_\pm$ nonnegative. The five stabilizer pairings and the unique nonzero-shell adjoint pairings then vanish. Every other entry in the exact collision ledger is off shell, while the nonzero-Fourier scalar complex and constant-twist cross column are boundedly removable. Theorem 6.1 therefore supplies a bounded correction. Necessity follows from the same complete obstruction ledger. $\square$

Because $\omega_- < \omega_e < \omega_+$, the interval (25) strictly contains the interval obtained when only the Einstein-plus primary is allowed to balance the negative branch. Thus the extra primary creates no new resonance at this fixture but does enlarge the bounded nonlinear cone. This is a theorem for one tuned $\ell = 2, m_A = 0, |k|$ fibre, not for nonaxisymmetric modes, other harmonics or circumferences, or multiple absolute-momentum fibres.

## 11 All-angular-momentum promotion and the two-momentum workload

### 11.1 The tuned standard-branch cone for every $\ell \geq 2$

The standard Einstein-minus calculation admits a uniform angular-momentum promotion. For each integer $\ell \geq 2$, tune the compact momentum by

$$k^2 = \sqrt{2\ell(\ell+1)} - \frac{\ell}{2} - \frac{1}{6}. \tag{28}$$

Let $a_\pm$ and $p_\pm$ denote the axial and polar Einstein-minus amplitudes at momenta $\pm k$. The complete standard-branch collision census has one relevant opposite-momentum characteristic shell. Up to nonzero exact normalizations $R_\ell > 0$ and $X_\ell \neq 0$, its two adjoint pairings are

$$\mathcal{R}_{\mathrm{polar}} = R_\ell \left( a_+ a_- - \frac{\ell(\ell+1)}{2} p_+ p_- \right), \tag{29}$$

$$\mathcal{R}_{\mathrm{axial}} = X_\ell \left( a_+ p_- - a_- p_+ \right). \tag{30}$$

Their complex zero variety consists of two mixed sheets

$$a_\pm = \sigma \sqrt{\frac{\ell(\ell+1)}{2}}\, p_\pm, \qquad \sigma \in \{+1, -1\}, \tag{31}$$

and the two one-sided planes on which one momentum sign vanishes.

**Theorem 11.1** (All-$\ell$ tuned axisymmetric standard cone)**.** *For every $\ell \geq 2$ at the tuning (28), the complete bounded quasiperiodic cone on the declared axisymmetric standard Einstein-minus/Einstein-plus carrier is the origin together with the two mixed sheets (31) satisfying*

$$\frac{1 - r_\ell}{1 + r_\ell} \leq \frac{|p_+|^2}{|p_-|^2} \leq \frac{1 + r_\ell}{1 - r_\ell}, \qquad r_\ell := \frac{\omega_-}{\omega_+}. \tag{32}$$

*The one-sided resonance planes meet the Hamiltonian and circle-momentum balance equations only at the origin.*

The proof is uniform in $\ell$: the exact collision census excludes every other standard-branch shell, equations (29)–(30) give the full resonance ideal, and the two global charge equations reduce to the elementary occupation interval (32). This is a genuine all-$\ell$ theorem, but it is still separately tuned at each $\ell$, axisymmetric, and confined to the standard branches. It does not join different angular momenta or different absolute compact momenta.

### 11.2 Two absolute momenta: a branch-basis obstruction census

The first two-absolute-momentum enlargement has closed all 164 declared branch-basis coefficients. The axisymmetric $L = 4$ calculation is complete: all 108 coefficients across axial–axial, polar–polar, and the two ordered cross-parity sectors are exact. The nonaxisymmetric $L = 3$ calculation contributes 44 exact coefficients and the $L = 1$ companion the remaining 12. Every declared individual branch-basis fixture in these two blocks has nonzero cokernel and a bounded obstruction.

This census by itself is only an exhaustive coefficient statement for the named basis columns: different columns may cancel after amplitudes are joined. The candidate-13 fibre is the first case where the arbitrary-amplitude ideal, the zero-frequency receiver, and the correction-class-dependent range have all been joined, giving the complete formula displayed after Theorem 6.1. The other collision circumferences admit a sharp scalar classification.

**Theorem 11.2** (Collision scalar-separation classification)**.** *On every positive-$\rho$ generic carrier with signed momenta $(1, -2)$, the common zero of $\mu_H, \mu_{P_x}, R_c$ is the origin. Consequently the complete bounded generic cone is $\{0\}$ at collision candidates (1) through (15). At each same-sign candidate (16) through (21), the scalar common zero is nontrivial and no strict separator in $\operatorname{span}(\mu_H, \mu_{P_x}, R_c)$ exists. The six full resonance-joined bounded cones are not classified by this scalar theorem.*

*Proof.* For the opposite-sign carrier let $(t_1, t_2)$ be the frequency midpoints between the Einstein-minus and extra branches at absolute momenta one and two. The shell polynomial associated with $(a, b, c) = (1, t_2/2 - t_1, -t_1 t_2/2)$ factors as

$$Q_1(\omega) = (\omega - t_1)(\omega + t_2/2), \qquad Q_{-2}(\omega) = (\omega - t_2)(\omega + 2t_1).$$

The second factors are positive. The first factors are negative on the negative-current Einstein-minus branch and positive on the extra and Einstein-plus branches. After the current sign is included, the resulting charge/pressure functional is a strictly positive sum of all absolute-current occupations. For each of the six same-sign collision values, the certificate instead gives four positive exact cofactors $y_i$ satisfying

$$\sum_i y_i s_i(\omega_i^2, n_i\omega_i, n_i^2) = 0.$$

Farkas duality excludes a strict scalar separator, while the same positive weights give a nonzero scalar common-zero occupation. Finite-frequency resonances are separate conditions and are not inferred to vanish. □

**Theorem 11.3** (Sharp collision nonemptiness split)**.** *On each of the six distinct same-sign collision backgrounds (16)–(21), the complete generic bounded second-order cone contains a nonzero real point. Together with the preceding theorem, the twenty-one collision fibres split sharply as*

$$\mathcal{Z}^{(i)}_{2,\mathrm{bounded}} = \{0\} \quad (1 \le i \le 15), \qquad \mathcal{Z}^{(i)}_{2,\mathrm{bounded}} \ne \{0\} \quad (16 \le i \le 21).$$

*This is a nonemptiness theorem for the last six cones, not a classification of their full real algebraic geometry.*

*Proof.* For each same-sign fibre, use the four positive exact Farkas cofactors as absolute-current occupations and take axisymmetric input, so all three lifted rotation moment maps vanish. The same cofactors set $\mu_H = \mu_{P_x} = R_c = 0$. All 108 nonzero-frequency same-fibre temporal channels are off shell: the exact census contains 864 nonzero target-shell defects, while its $L = 0$ rows use the separately empty nonzero-Fourier and homogeneous nonzero-frequency quotients.

It remains only to kill the one cross-fibre resonance assigned to each candidate. On (17)–(19), the Farkas support omits one required resonant factor. On (16) and (20), choose the occupied modes axial with $m = 0$; the relevant $L = 3$ or $L = 1$ coefficient contains $\langle 2, 0; 2, 0 \mid L, 0\rangle = 0$. On (21), put the resonant $q_+(1), q_-(2)$ pair on the certified real mixed-parity $L = 4$ component $A_{\mathrm{pol}} = rA_{\mathrm{ax}}$, $B_{\mathrm{pol}} = sB_{\mathrm{ax}}$. Its two real nonzero parity vectors have positive absolute-current norms and may be scaled independently to the prescribed occupations. The complete finite-harmonic cokernel criterion then supplies a bounded finite-quasiperiodic second-order correction on each of the six backgrounds. □

**Corollary 11.4** (Universal scalar cone and bounded section)**.** *For every $\rho > 0$ on a same-sign $n = (1, 2)$ carrier, the nonnegative $(\mu_H, \mu_{P_x}, R_c)$-null occupation cone has exactly four extreme rays. Each contains both Einstein-minus nodes and chooses one of $p_{\mathrm{extra}}, q_+$ independently on each momentum fibre. On each of the six collision backgrounds (16)–(21), all four scalar extreme rays* *admit a nonzero bounded amplitude lift. More strongly, every point of each complete four-ray scalar cone has a bounded amplitude lift. Thus the bounded cone projects surjectively onto the scalar occupation cone on all six backgrounds.*

*Proof.* Divide the scalar receiver column on branch $b$ and fibre $n$ by the positive factor $n^2$, and put $x_{b,n} = \omega_{b,n}/n = \sqrt{\rho + \mu_b/n^2}$. The columns lie on $(x^2, x, 1)$, with $\rho$-independent order

$$q_-^{(2)} < p^{(2)} < q_+^{(2)} < q_-^{(1)} < p^{(1)} < q_+^{(1)}.$$

The only non-obvious gap is $(9 - 5\sqrt{3})/2 > 0$, which follows from $81 > 75$. The current signs in this order are $(-, +, +, -, +, +)$. Four ordered moment-curve columns have the unique dependence

$$z_i = \frac{1}{\prod_{j \neq i}(x_i - x_j)}, \qquad \sum_i z_i x_i^d = 0 \quad (d = 0, 1, 2),$$

whose signs alternate. Hence exactly four supports agree with the current signs, proving the extreme-ray classification.

For the 24 lifts, ten supports omit a resonant factor and ten use the axisymmetric odd-$L$ zero. The remaining four use real $L = 4$ components: two on candidate (19)'s regular-pencil eigenline and two on candidate (21)'s mixed-parity component. Independent fibre rescaling matches the extreme-ray occupations. This proves ray saturation only; cross terms between different ray lifts can reactivate the bilinear resonance map.

For occupation-surjectivity, do not add fixed ray representatives. Instead, given an arbitrary point of the scalar cone, choose all occupied vectors axial and axisymmetric on candidates (16), (17), (18), and (20). The odd- $L$ coefficient then vanishes independently of the six occupation values. On (19) and (21), place the two resonant vectors on one fixed real mixed $L = 4$ component and rescale the two fibres independently to their prescribed absolute-current norms; choose all spectator vectors axial with $m = 0$. This defines a section on every cone face, including pairwise ray sums and the origin. It does not imply that every amplitude in the fibre over a given occupation is bounded. □

**Proposition 11.5** (Exact same-sign bounded fibre products)**.** *For each candidate $i = 16, \ldots, 21$, let $\pi_i$ be the six-node absolute-current occupation map, $C_i$ its four-ray $(\mu_H, \mu_{P_x}, R_c)$-zero cone, $\mu_J$ the three lifted rotation moment maps, and $B_i$ the complete isolated cross-fibre resonance map. Then the bounded or finite-quasiperiodic second-order tangent cone is exactly*

$$\mathcal{Z}_i^{\text{bnd}} = \pi_i^{-1}(C_i) \cap \mu_J^{-1}(0) \cap \mathcal{V}(B_i).$$

*All relative phases, both parities and all $m = -2, \ldots, 2$ coefficients are retained.*

*Proof.* Necessity is the complete finite-harmonic adjoint-cokernel theorem: the scalar factor gives $H, P_x, R_c$, the rotation factor gives the three lifted rotations, and $\mathcal{V}(B_i)$ kills the unique candidate-specific cross-fibre shell. Conversely, the 864-defect same-fibre census makes every other nonzero-frequency source block invertible, so simultaneous vanishing is sufficient. This gives an exact equation, not yet a decomposition of its real Hermitian components or singular strata. □

**Proposition 11.6** (Resonance fibres over the scalar-cone faces)**.** *On each same-sign collision background (16)–(21), the complete complex positive-frequency phase/parity resonance fibre is stratified over the four-ray scalar cone. The two $q_-$ occupations are strictly positive at every nonzero* *scalar-cone point. The automatic and active strata are*

| *candidate* | *resonant pair* | *automatic face* | $\dim_{\mathbb{C}} \mathcal{V}_{\text{active}}$ | $\# \operatorname{Irr}_{\mathbb{C}} \mathcal{V}_{\text{active}}$ |
|---|---|---|---|---|
| 16 | $q_-^{(1)} q_-^{(2)}$ | $\{0\}$ | 12 | 1 |
| 17 | $q_-^{(1)} q_+^{(2)}$ | $\operatorname{cone}(R_1, R_2)$ | 14 | 1 |
| 18 | $p^{(1)} q_-^{(2)}$ | $\operatorname{cone}(R_2, R_4)$ | 22 | 1 |
| 19 | $p^{(1)} q_-^{(2)}$ | $\operatorname{cone}(R_2, R_4)$ | 10 | 4 |
| 20 | $q_+^{(1)} q_-^{(2)}$ | $\operatorname{cone}(R_1, R_3)$ | 14 | 1 |
| 21 | $q_+^{(1)} q_-^{(2)}$ | $\operatorname{cone}(R_1, R_3)$ | 10 | 2 |

*Candidates (17) and (20) use their isolated difference-frequency channel; the other four rows use the isolated sum-frequency channel. On an automatic face one bilinear factor vanishes. Off it, the table lists the complete active all-m complex resonance variety. The four components on (19) are the real-supported regular-pencil eigenline components, and the two on (21) are the real mixed parity-proportionality components. Every scalar-cone point has at least one real representative in the corresponding stratum.*

*Proof.* Every extreme ray contains $q_-^{(1)}$ and $q_-^{(2)}$ with positive weight. Reading the optional resonant node from the four ray supports gives the automatic faces in the table. On their complements, restrict the complete binary-form zero-variety decompositions: the target-doublet first-transvectant variety for (16), the third-transvectant kernels for (17),(20), the multiplicity-two first-transvectant variety for (18), the regular pencil for (19), and the scalar highest-spin product for (21). Their certified dimensions and component counts give the last two columns. The scalar-cone section in the preceding corollary supplies real points. Combined with the exact fibre-product formula, these strata are intersected with the prescribed real norm levels and the three lifted rotation moment maps to give a necessary-and-sufficient bounded equation. The proposition does not decompose that real intersection into connected components or singular strata. □

**Proposition 11.7** (Connected automatic-face rotation links)**.** *For candidates (17)–(21), fix a nonzero occupation in any relative support stratum of the automatic two-ray face in the preceding table. The complete amplitude link satisfying*

$$B_i = 0, \qquad \mu_{J_1} = \mu_{J_2} = \mu_{J_3} = 0$$

*is nonempty and connected. Candidate (16) has no nonzero automatic face, so this statement is not applicable there. This is a fixed-occupation theorem; it neither glues distinct occupation strata nor classifies the active resonance varieties.*

*Proof.* On an automatic face, one complete factor of the bilinear map $B_i$ vanishes. For each occupied branch–momentum node, fix its positive absolute-current norm. The amplitude space is then a product of odd spheres. Quotienting by the independent node phases gives a compact connected product of complex projective spaces. The action-derived Lee–Wald form descends to a signed product of Fubini–Study forms. A negative $q_-$ factor reverses orientation but remains nondegenerate.

The diagonal lifted $SO(3)$ action is Hamiltonian, and its moment map is the descended triple $(\mu_{J_1}, \mu_{J_2}, \mu_{J_3})$. Compactness makes this moment map proper. The connected-fibre theorem of Lerman, Meinrenken, Tolman and Woodward [9] therefore makes its zero fibre connected. The certified axisymmetric representative proves nonemptiness. Finally, its inverse image in the product of norm spheres has the connected node-phase torus as fibre, and is therefore connected. □

**Proposition 11.8** (The axisymmetric section is rotation-critical)**.** *On the certified all-$m = 0$ section over every scalar-cone point of each candidate (16)–(21), the Jacobian of the three lifted rotation moment maps has rank zero at the origin and exactly two at every nonzero point. Hence zero is not a regular value along this section, and these explicit bounded points cannot seed a codimension-three implicit-function chart.*

*Proof.* In the spin-two basis $m = -2, -1, 0, 1, 2$, let $e_0$ denote the $m = 0$ vector. The exact invariant angular form gives

$$T_3 e_0 = 0, \qquad W_2 T_1 e_0 \text{ and } W_2 T_2 e_0 \quad \text{real-linearly independent.}$$

Thus $d\mu_{J_3} = 0$ on every occupied axisymmetric block, giving rank at most two. At a nonzero section point, at least one block has nonzero norm; variations supported in that block retain the two independent $T_1, T_2$ covectors with a nonzero current coefficient, giving rank at least two. At the origin all three differentials vanish because the moment map is quadratic. In particular, the connected automatic-face links above may contain critical points: connectedness does not imply regularity. The next local gate is the quadratic normal form of the missing $J_3$ equation on the kernel of the two transverse differentials. □

**Proposition 11.9** (Automatic-face hyperbolic normal form)**.** *Fix a nonzero axisymmetric section point on any automatic-face support stratum of candidates (17)–(21), and let $N$ be the number of occupied current eigenlines. On the angular slice aligned with those eigenlines, the quadratic $\mu_{J_3}$ form restricted to*

$$\ker d(\mu_{J_1}, \mu_{J_2})$$

*has real inertia*

$$(n_+, n_-, n_0) = (4N - 2, 4N - 2, 2).$$

*In particular it is indefinite. Moreover, through every such point there is an exact fixed-occupation nonaxisymmetric curve inside the complete bounded fibre product. Candidate (16) has no nonzero automatic face, so the statement is not applicable there.*

*Proof.* The spin-two angular form and $J_3$ form in the ordered basis $m = -2, -1, 0, 1, 2$ are

$$W_2 = \operatorname{diag}\left(1, \frac{1}{4}, \frac{1}{6}, \frac{1}{4}, 1\right), \qquad W_2 T_3 = \operatorname{diag}\left(-2, -\frac{1}{4}, 0, \frac{1}{4}, 2\right).$$

Rescale each node by the square root of its nonzero current magnitude and swap the $m = +1$ and $m = -1$ variables on negative-current nodes. This normalizes all coefficients simultaneously. Put $u_j = \delta z_{j,+1}$ and $v_j = \overline{\delta z_{j,-1}}$. The two real transverse rotation equations combine into one complex equation

$$\sum_{j=1}^{N} a_j (u_j + v_j) = 0, \qquad a_j > 0.$$

With $w_j = u_j + v_j$ and $z_j = u_j - v_j$, the $m = \pm 1$ part of $\mu_{J_3}$ is proportional to $\operatorname{Re} \sum_j w_j \overline{z_j}$. The constraint places $w$ in $a^\perp$. Its pairing with the $a^\perp$ part of $z$ gives real inertia $(2N - 2, 2N - 2)$, while the complex component of $z$ parallel to $a$ is a two-real-dimensional radical. The unrestricted $m = \pm 2$ coordinates contribute $(2N, 2N, 0)$, proving the displayed inertia.

For the exact curve, choose any occupied eigenline, fix its phase so that its $m = 0$ coefficient is $a > 0$, and replace its angular vector by

$$z(t) = \sqrt{a^2 - 12t^2}\, e_0 + te_{+2} + te_{-2}, \qquad 12t^2 < a^2.$$

Its norm is fixed because $(a^2 - 12t^2)/6 + t^2 + t^2 = a^2/6$. No adjacent magnetic weights are simultaneously occupied, so $\mu_{J_1} = \mu_{J_2} = 0$, while the two endpoint terms cancel in $\mu_{J_3}$. The absent resonant node stays zero on an automatic face, hence $B_i = 0$ remains exact. This proves the curve lies in the necessary-and-sufficient bounded fibre product. The proposition does not classify internal current-orthogonal directions or the two-dimensional radical, and therefore is not a complete singular-stratum theorem. □

**Corollary 11.10** (Complete fixed-norm rotation Hessian)**.** *At the same axisymmetric points, let $N$ be the number of occupied scalar branch–momentum nodes and let $D$ be the sum of their complex internal current-space dimensions. A $q_-$ or $q_+$ node has $D_j = 2$, one axial and one polar current eigenline, whereas a p-extra node has $D_j = 4$, two eigenlines in each parity. On the complete tangent to the fixed-node-norm link, restricted by $\mathrm{d}(\mu_{J_1}, \mu_{J_2}) = 0$, the real inertia of the quadratic $\mu_{J_3}$ form is*

$$(4D - 2, 4D - 2, 2D - N + 2).$$

*After quotienting the $N$ independent node phases it is*

$$(4D - 2, 4D - 2, 2D - 2N + 2).$$

*Thus every ray and two-ray relative-interior stratum of the five nonzero automatic faces remains a transverse rotation saddle after all axial and polar internal directions are restored. Candidate* (16) *remains not applicable.*

*Proof.* The aligned block is the preceding proposition. A current-orthogonal internal eigenline does not enter the two linear transverse-rotation equations. Its five complex angular coefficients see the exact weighted $J_3$ diagonal

$$\left(-2, -\frac{1}{4}, 0, \frac{1}{4}, 2\right),$$

and hence add real inertia $(4, 4, 2)$. Summing over the $D - N$ orthogonal eigenlines gives angular-kernel inertia $(4D - 2, 4D - 2, 2)$. The $m = 0$ fixed-node-norm tangent is radical of real dimension $2D - N$, which yields the first formula. The node-phase torus lies in that radical and removes $N$ real directions, yielding the second. The dimensions are respectively $10D - N - 2$ and $10D - 2N - 2$, equal to the sums of the displayed inertia triples. Exact enumeration of all fifteen ray/interior support strata gives phase-reduced inertias ranging from $(30, 30, 10)$ to $(54, 54, 20)$.

This is a complete quadratic Hessian theorem at fixed node norms, not a classification of the nonlinear zero set along its radical. It neither glues occupation strata nor treats active resonance components. □

**Proposition 11.11** (Candidate-(16) active restricted current)**.** *Fix both nonzero $q_-^{(1)}$ and $q_-^{(2)}$ current norms on candidate (16), and quotient their independent phases. The irreducible active resonance variety is a complex projective tenfold in $\mathbb{CP}^9 \times \mathbb{CP}^9$. On every complex smooth stratum, the restricted Lee–Wald form is nondegenerate and negative Kähler. On the generic smooth locus its real symplectic rank is* 20*.*

*Proof.* Before projectivization, the exact target-doublet $L = 3$ resonance equations factor into two rank-at-most-one determinantal cones. Their product is irreducible of complex dimension 12 in the 20-dimensional affine input space. The equations are bihomogeneous in the two nonzero nodes, so quotienting their two complex scalings gives the stated projective dimension.

Both resonant nodes are on the $q_-$ branch. The complete axial/polar current is therefore negative definite on each node. Writing its descended Hermitian form as

$$h = -c_1 H_1 \oplus -c_2 H_2, \qquad c_1, c_2 > 0,$$

gives $h(v, v) < 0$ for every nonzero complex tangent vector on every smooth stratum. Its imaginary part is consequently nondegenerate there. The projective variety itself is singular, so this proposition does not apply a smooth-orbifold connected-fibre theorem to its full rotation-zero set. □

**Proposition 11.12** (Active linear-sheet rotation links). *Each of the four real pencil-eigenline components on candidate (19) and the two real parity-proportional components on candidate (21) has a nondegenerate restricted Lee–Wald current. Its resonant core has Hermitian inertia* $(5, 5, 0)$. *At every fixed nonzero active occupation, each component has a nonempty connected lifted-rotation zero link, including arbitrary occupied spectator nodes. The four and two components remain distinct.*

*Proof.* Every declared sheet is the direct sum of one complex five-dimensional subspace of a positive-current node and one complex five-dimensional subspace of a negative-current node. Each ambient node form is definite, so its restriction to a nonzero linear graph or pencil eigenline remains definite. Spectral orthogonality between distinct branch–momentum nodes then gives inertia $(5, 5, 0)$; occupied spectators add definite orthogonal blocks and cannot create a radical.

After fixing the two resonant-node norms and quotienting their phases, the core is $\mathbb{CP}^4 \times \mathbb{CP}^4$ with one positive and one negative weighted Fubini–Study factor. Spectators add projective factors. The diagonal lifted $SO(3)$ action preserves each internal graph or eigenline and is Hamiltonian. Its zero fibre is nonempty by choosing the spin-two $m = 0$ vector in every factor. Compactness makes the moment map proper, so the connected-fibre theorem gives a connected zero fibre; restoring the connected node-phase torus preserves connectedness. Distinct parity ratios and pencil eigenlines are unchanged by node phases, and the active nonzero-norm condition prevents passage through a one-sided component. □

**Proposition 11.13** (Candidate-(17)/(20) axisymmetric Zariski currents). *At every nonzero active scalar-cone point on the certified all-axial* $m = 0$ *sections of candidates (17) and (20), the affine Zariski tangent to the complete third-transvectant resonance variety has complex dimension* 16 *and restricted Hermitian current inertia*

$$(6, 10, 0).$$

*The section is algebraically singular because the irreducible variety has complex dimension* 14. *Nevertheless, the restricted current is nondegenerate. After removing the two node scalings, the projective Zariski tangent has inertia* $(5, 9, 0)$ *and real symplectic rank* 28.

*Proof.* For the axisymmetric binary quartic $e_0 = (0, 0, 1, 0, 0)$, the exact third-transvectant matrix has rank two and a three-dimensional kernel. The two linearized parity equations therefore have total rank four in the twenty-dimensional affine input space, giving the stated tangent dimension.

The only possible current radical is controlled by equality of the absolute currents on the positive and negative resonant nodes. Exact outward rational intervals prove the strict opposite inequality on both active extreme rays of each scalar cone:

$$\mathsf{n}(q_-^{(1)}) > \mathsf{n}(q_+^{(2)}) \quad \text{on candidate (17)}, \qquad \mathsf{n}(q_-^{(2)}) > \mathsf{n}(q_+^{(1)}) \quad \text{on candidate (20)}.$$

Every inactive ray adds only to the negative occupation, so the inequalities hold throughout the active cones. In each parity channel, the three kernel directions give three positive/negative pairs and the two constrained directions are negative, yielding $(3, 5, 0)$. Doubling parity gives $(6, 10, 0)$. The two complex node-scaling directions have inertia $(1, 1, 0)$, and their removal gives $(5, 9, 0)$.

This is a Zariski-tangent theorem at singular points. It does not classify the restricted current on the full smooth locus or justify a connected-fibre argument there. $\square$

**Proposition 11.14** (Smooth current radicals on candidates (17) and (20))**.** *Each complete active third-transvectant variety contains a smooth bounded point, with all five stabilizer moment maps zero, at which the fixed-norm projective Lee–Wald current has a nonzero radical. Consequently neither active variety is globally symplectic, and the smooth symplectic-orbifold connected-fibre theorem cannot be applied globally.*

*Proof.* The exact parity pencil has coefficient matrix $C_0 = \left(\begin{smallmatrix} a & b \\ b & 3a \end{smallmatrix}\right)$, with $b/(a\lambda) = \epsilon\sqrt{3}$, $\epsilon = \pm 1$, and $\lambda^2 = 128/5$. Put

$$D = \operatorname{diag}(1, 3), \qquad Q = \begin{pmatrix} \epsilon\sqrt{3} & -\epsilon\sqrt{3} \\ 1 & 1 \end{pmatrix}, \qquad P^T = (C_0 Q)^{-1}.$$

Then $Q^T D Q = 6I$, while

$$P = Q \operatorname{diag}\left(\frac{1}{6a(\lambda+1)}, -\frac{1}{6a(\lambda-1)}\right).$$

Thus the second transform is not a scalar multiple of the first. After the independent channel rescaling

$$S = \operatorname{diag}\left(\frac{3}{2}a(\lambda+1), \frac{3}{2}a(\lambda-1)\right), \qquad PS = Q \operatorname{diag}(1/4, -1/4),$$

both eigenchannels have normalized positive-to-negative current coefficient ratio $1/16$.

In either channel take

$$f = (1, 0, 0, 0, 1), \qquad g = (1, 0, 1, 0, 1).$$

The third-transvectant equations vanish and their Jacobian has rank three, so this is a smooth point with seven-dimensional affine tangent. Restricting $-H \oplus H/16$, where $H = \operatorname{diag}(1, 1/4, 1/6, 1/4, 1)$, gives rank six and the exact radical

$$\delta f = (0, 1/4, 0, 1/4, 0), \qquad \delta g = (0, 1, 0, 1, 0).$$

It is tangent to both norm levels and orthogonal to both phase directions, so it survives projectivization. Since $H(f, f) = 2$ and $H(g, g) = 13/6$, its positive-to-negative occupation ratio is $13/192$. Exact positive mixtures $R_3 + sR_1$ on candidate (17) and $R_2 + sR_1$ on candidate (20) realize that ratio inside their scalar cones. The reflection-symmetric representatives and their spectators have zero angular moments, while the scalar-cone equations kill the time and circle moments. The exact bounded fibre-product theorem then supplies the stated bounded points.

This is an existence theorem for a smooth degeneracy locus, not a classification of its divisor or presymplectic quotient. $\square$

**Proposition 11.15** (Smooth current radicals on candidate (18))**.** *The complete candidate-(18) active resonance variety contains two exact families of smooth bounded points with all five stabilizer moment maps zero. On either family, the fixed-norm projective Lee–Wald current has a four-complex-dimensional radical. Hence this active variety is not globally symplectic either.*

*Proof.* Write $X, Y$ for the current-orthogonal active functionals on the axial and polar $p$-extra doublets and $U, V$ for the axial and polar $q_-$ amplitudes. The exact $L = 3$ source pencil is equivalent to

$$T_1(X,U) + T_1(Y,V) = 0, \qquad 3T_1(X,V) + T_1(Y,U) = 0.$$

With

$$Q = \begin{pmatrix} \sqrt{3} & -\sqrt{3} \\ 1 & 1 \end{pmatrix}, \qquad P = Q^{-T},$$

one has $P^T Q = I$ and $P^T \left(\begin{smallmatrix} 0 & 3 \\ 1 & 0 \end{smallmatrix}\right) Q = \mathrm{diag}(\sqrt{3}, -\sqrt{3})$. The resonance equations therefore factor into two rank-one conditions $T_1(f_+, g_+) = T_1(f_-, g_-) = 0$.

Let $w_x, w_y > 0$ be the exact quotient-current weights of $X, Y$, and let $h_- > 0$ be the absolute axial $q_-$ weight. Direct reduction of the action-derived current gives

$$G_p = \begin{pmatrix} w_x/12 + w_y/4 & -w_x/12 + w_y/4 \\ -w_x/12 + w_y/4 & w_x/12 + w_y/4 \end{pmatrix}, qquad |G_-| = 6h_- I.$$

Thus $z_+ = (1,1)$ and $z_- = (1,-1)$ are exact eigenlines of $G_p$, with eigenvalues $w_y/2$ and $w_x/6$, respectively. On either line set $f = z_\pm \otimes e_0$, $g = t_\pm z_\pm \otimes e_0$, where $e_0 = (0,0,1,0,0)$ and

$$t_+^2 = \frac{w_y/2}{6h_-}, qquad t_-^2 = \frac{w_x/6}{6h_-}.$$

Both entries of $z_\pm$ are nonzero, so both rank-one factors are at smooth nonzero proportional pairs. For every $r \perp e_0$,

$$\delta f = z_\pm \otimes r, \qquad \delta g = t_\pm z_\pm \otimes r$$

is tangent and current-orthogonal to the complete tangent. These four complex directions are tangent to both norm levels and transverse to the two phase directions, so they survive projectivization.

Finally, exact rational intervals put the active occupation ratio below one on $R_1$ and above one on $R_3$. The unique positive mixture $R_3 + s_{18} R_1$ therefore gives equal $p_{\mathrm{extra}}^{(1)}$ and $q_-^{(2)}$ absolute-current occupations, precisely the scaling above. The $m = 0$ resonant carriers and spectators kill the three rotation moments, while the scalar cone kills the time and circle moments. The exact bounded fibre-product theorem completes the claim.

This proves two smooth degeneracy families, not their complete divisor or presymplectic quotient. □

## 12 Current bounded frontier

The electric, Wilson, and circumference columns are closed, and the first full-time velocity repair sharpens the $a, d$-frontier. Earlier direct $d$-source fixtures were evaluated at $t = 0$. They correctly proved that the constant shell-projection matrix is invertible, but they could not prove that the full source was bounded. No frozen paper theorem asserted the stronger conclusion; the intermediate "complete $d$-column" reading is withdrawn and replaced by the following full-time statement.

Locality gives

$$\mathcal{E}^{(2)}[dt, u](t) = t\, \mathcal{E}^{(2)}[c, u](t) + \mathcal{E}^{(2)}[dt, u](0).$$

For the two axial extra representatives and the first polar representative, the coefficient of $t$ vanishes. For the second polar representative it is the nonzero eight-row vector

$$(-36, 0, 164, -8\sqrt{3}i, 0, -100, -24, -20).$$

Thus the isolated fixture has polynomial zero locus $dz_{\mathrm{polar},2} = 0$. The previously computed constant-shell adjoint matrix remains valid, but cannot by itself establish boundedness. Combining this restored term with the four direct radion-position columns gives the complete positive-degree cross ideal on the $\ell = 2, k = 0$ extra multiplet:

$$\langle az_{\mathrm{ax},1}, az_{\mathrm{ax},2}, az_{\mathrm{pol},1}, az_{\mathrm{pol},2}, dz_{\mathrm{pol},2}\rangle. \tag{33}$$

These five monomials are already a Gröbner basis in the declared amplitude order. Their zero locus has three faces:

1. all extra amplitudes zero, with $a, d$ unrestricted by this cross ledger;
2. $a = 0$ and $z_{\mathrm{pol},2} = 0$, with $d$ and the two axial plus first polar amplitudes free;
3. $a = d = 0$, with all four extra amplitudes free.

This is a complete $\mathcal{P}$-zero locus for the declared cross block, not by itself a complete bounded cone. Before the other conditions are imposed, its surviving faces must still satisfy moment maps, constant shell pairings, self-products, and twist cross terms. On the complete global-plus-extra carrier those remaining conditions now close as follows.

**Theorem 12.1** (Complete global plus $\ell = 2, k = 0$ extra bounded cone)**.** *Adjoin all four axial/polar $\ell = 2, k = 0$ extra-primary amplitudes $(x_{\mathrm{ax},1}, x_{\mathrm{ax},2}, x_{\mathrm{pol},1}, x_{\mathrm{pol},2})$ to the complete homogeneous and twist carrier. The $\mathscr{X}_{\mathrm{qp}}$ tangent cone is exactly*

$$\mathcal{Z}^{\mathrm{qp}}_{2,\,\mathrm{global}+\ell 2X} = \{(c, d, W_x, A) : c, d, W_x \in \mathbb{R},\ A \in \mathbb{R}^3\}. \tag{34}$$

*In particular, it contains no nonzero $\ell = 2, k = 0$ extra-primary direction.*

*Proof.* The universal polynomial rows first force $b = B = 0$ and $Q_e a = 0$. The remaining Hamiltonian moment map is

$$\mu_H = -a^2 - Q_e^2 - \frac{4}{3}X, \tag{35}$$

$$X = 1296|x_{\mathrm{ax},1}|^2 + \frac{208}{3}|x_{\mathrm{ax},2}|^2 + 22464|x_{\mathrm{pol},1}|^2 + 12288|x_{\mathrm{pol},2}|^2. \tag{36}$$

It is a strictly negative sum of squares, so its real zero locus has $a = Q_e = x_{\mathrm{ax},1} = x_{\mathrm{ax},2} = x_{\mathrm{pol},1} = x_{\mathrm{pol},2} = 0$. What remains is precisely the standard cone of Theorem 7.1, whose bounded correction is already explicit. The repaired ideal (33) is automatically satisfied there. □

After these repairs, the still-open full-carrier work is:

1. standard Einstein oscillators and exceptional $\ell = 1$ inputs outside the separately classified fixtures, other harmonics, and generic nonzero momenta;
2. their $a, d$ polynomial columns and shell-pairing zero loci;
3. shell resonances among bounded wave products and remaining mixed blocks; and
4. the common zero locus for arbitrary finite sums.

Any future completion must preserve the separation between positive-degree obstructions $\mathcal{P}$ and shell pairings $\mathcal{R}$.

| Carrier or column | $\mathscr{X}_{\rm ep}$ correction | $\mathscr{X}_{\rm qp}$ correction |
|---|---|---|
| Complete finite-support cone | iff five moment maps vanish | additional $\mathcal{P}$ and $\mathcal{R}$ conditions |
| Standard generalized-zero block | classified | classified completely |
| Global plus $\ell = 2, k = 0$ extra | moment-map cone contains balanced extra directions | bounded cone equals $(c, d, W_x, A)$; no extra wave |
| Electric charge $\times$ oscillator | finite quasiperiodic | removable by duality transport |
| Wilson $\times$ oscillator | zero source | zero source |
| Circumference $\times$ $k = 0$ mode | ordinary same-frequency | removable |
| Circumference $\times$ $k \neq 0$ mode | secular radius derivative | obstructed |
| Velocity $d\times$ selected polar extra mode | finite polynomial source | nonzero $\mathcal{P}_{j,1}$ unless canceled jointly |
| Exceptional axial–polar interior | secular primitive on moment cone | bounded-resonance obstructed |
| Tuned $\ell = 2, m_A = 0$, nonzero $\lvert k\rvert$, all primaries | moment-map cone | Theorem 10.1: two mixed sheets over the exact occupation polytope |
| $a, d \times \ell = 2, k = 0$ extra | allowed on moment cone | polynomial ideal (33), then $\mathcal{R}$ still required |
| Remaining $a, d$ oscillator columns | allowed by ep theorem on moment cone | open |

## 13 Interpretation

The solution set near a symmetric compact background is not generally a linear space. Its second-order approximation is a cone cut out first by a moment map and then, if uniform boundedness is demanded, by resonance conditions. Three points are important.

First, a linearly nonradical extra Weyl mode may fail to be tangent to the nonlinear fixed-charge solution space. This is nonlinear selection, not linear gauge removal. Second, mixing Einstein and extra directions can cancel the moment maps, so no conclusion follows from the sign of a pure branch alone. Third, even complete moment-map cancellation leaves bounded resonances. Exponential-polynomial formal continuation and bounded perturbative stability are therefore different claims.

The tuned nonzero-momentum theorem adds a complementary point: enlarging the positive branch can widen a surviving bounded cone without introducing a new shell collision. The extra primary is therefore neither uniformly deleted nor uniformly liberated by the quadratic test; its disposition depends on the complete carrier and correction class.

The result supplies the middle map in the programme's residual atlas:

$$X_\alpha \longmapsto \Omega(X_\alpha, \overline{X}_\alpha) \longmapsto \mu(X_\alpha) \longmapsto D^2\mathcal{E}[X_\alpha, X_\alpha].$$

Interaction sourcing, detector response, and BRST status are separate later maps.

## 14 What is not proved

The following remain outside the theorem:

- infinite harmonic sums and Sobolev completion;
- retarded or advanced solvability and nonlinear causal dependence;
- integration of a second-order jet to an exact or all-orders family;
- the final residual/background-stabilizer quotient;
- asymptotically flat, de Sitter, anti-de Sitter, or black-hole boundary conditions;
- observer coupling, particles, quantum BRST cohomology, and unitarity.

These exclusions are not cosmetic. In particular, Theorem 5.2 does not establish nonlinear stability.

# 15 Certificate and reproduction ledger

The manuscript is an OUTLINE-ALLOWED/DRAFT-ALLOWED integration of committed certificates. Its current source baseline is repository commit `d6d58957f899c1433f43c1c37d1618eb298ffb86`.

| Claim | Authoritative certificate |
|---|---|
| Complete finite-support exponential-polynomial cone and bounded ledger | `bridge/certificates/einstein_maxwell_weyl_complete_finite_harmonic_smooth_global_second_order.json` |
| Finite-generic bounded zero-frequency receiver | `bridge/certificates/einstein_maxwell_weyl_finite_generic_bounded_zero_block.json` |
| Complete candidate-13 bounded and smooth cone formulas | `bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_candidate13_complete_mixed_cone.json` |
| Candidate-13 scalar-separation bounded-origin theorem | `bridge/certificates/einstein_maxwell_weyl_candidate13_scalar_separation_no_go.json` |
| All-collision scalar-separation classification | `bridge/certificates/einstein_maxwell_weyl_collision_scalar_separation_classification.json` |
| Same-sign collision same-fibre shell census | `bridge/certificates/einstein_maxwell_weyl_same_sign_collision_same_fibre_census.json` |
| Six same-sign collision bounded witnesses | `bridge/certificates/einstein_maxwell_weyl_same_sign_collision_bounded_witnesses.json` |
| Universal same-sign scalar extreme rays | `bridge/certificates/einstein_maxwell_weyl_same_sign_scalar_extreme_rays.json` |
| Candidatewise 120-minor/90-circuit scalar audit | `bridge/certificates/einstein_maxwell_weyl_same_sign_collision_scalar_occupation_cones.json` |
| All 24 same-sign scalar extreme-ray lifts | `bridge/certificates/einstein_maxwell_weyl_same_sign_extreme_ray_lifts.json` |
| Bounded sections over the complete same-sign scalar cones | `bridge/certificates/einstein_maxwell_weyl_same_sign_scalar_cone_sections.json` |
| Exact same-sign bounded phase/parity fibre products | `bridge/certificates/einstein_maxwell_weyl_same_sign_phase_parity_fibre_product.json` |
| Complete same-sign resonance fibres over scalar-cone faces | `bridge/certificates/einstein_maxwell_weyl_same_sign_resonance_face_fibres.json` |
| Connected fixed-occupation rotation-zero links on automatic faces | `bridge/certificates/einstein_maxwell_weyl_same_sign_automatic_face_rotation_links.json` |
| Axisymmetric rotation critical-locus theorem | `bridge/certificates/einstein_maxwell_weyl_same_sign_axisymmetric_rotation_singularity.json` |

Continued on next page

| Claim | Authoritative certificate |
|---|---|
| Automatic-face aligned rotation normal form and exact arcs | `bridge/certificates/einstein_maxwell_weyl_same_sign_automatic_face_rotation_normal_form.json` |
| Complete phase-reduced internal rotation normal form | `bridge/certificates/einstein_maxwell_weyl_same_sign_automatic_face_full_internal_rotation_normal_form.json` |
| Complete unquotiented and phase-quotiented fixed-norm rotation Hessians | `bridge/certificates/einstein_maxwell_weyl_same_sign_automatic_face_full_rotation_normal_form.json` |
| Candidate-(16) active restricted-current theorem | `bridge/certificates/einstein_maxwell_weyl_same_sign_candidate16_active_restricted_current.json` |
| Candidate-(19)/(21) active linear-sheet rotation links | `bridge/certificates/einstein_maxwell_weyl_same_sign_active_linear_sheet_rotation_links.json` |
| Candidate-(17)/(20) axisymmetric Zariski-tangent currents | `bridge/certificates/einstein_maxwell_weyl_same_sign_candidate17_20_axisymmetric_restricted_current.json` |
| Candidate-(17)/(20) smooth active-current radicals | `bridge/certificates/einstein_maxwell_weyl_same_sign_L1_active_restricted_current_degeneracy.json` |
| Candidate-(18) smooth active-current radical families | `bridge/certificates/einstein_maxwell_weyl_same_sign_candidate18_active_restricted_current_degeneracy.json` |
| Stabilizer action and lifted rotations | `bridge/certificates/einstein_maxwell_weyl_plebanski_hacyan_stabilizer.json` |
| Generalized-mode symplectic form | `bridge/certificates/einstein_maxwell_exceptional_global_symplectic.json` |
| Generalized-mode moment maps | `bridge/certificates/einstein_maxwell_weyl_exceptional_global_moment_maps.json` |
| Standard generalized-zero bounded cone | `bridge/certificates/einstein_maxwell_weyl_standard_global_bounded_second_order.json` |
| Electric and Wilson transport | `bridge/certificates/einstein_maxwell_weyl_electric_wilson_complete_oscillator_transport.json` |
| Complete circumference column | `bridge/certificates/einstein_maxwell_weyl_circumference_complete_oscillator_bounded_classification.json` |
| First full-time velocity polynomial column | `bridge/certificates/einstein_maxwell_weyl_d_ell2_extra_full_time_polynomial.json` |
| Repaired $a, d \times \ell = 2$ polynomial zero locus | `bridge/certificates/einstein_maxwell_weyl_ad_ell2_extra_polynomial_zero_locus.json` |
| Complete global plus $\ell = 2, k = 0$ extra bounded cone | `bridge/certificates/einstein_maxwell_weyl_complete_global_ell2_extra_bounded_cone.json` |
| Exceptional two-polarization resonance | `bridge/certificates/einstein_maxwell_weyl_exceptional_ell1_two_polarization_resonance.json` |
| Opposite-momentum exponential-polynomial theorem | `bridge/certificates/einstein_maxwell_weyl_opposite_momentum_smooth_global_second_order.json` |
| Tuned axisymmetric all-primary bounded cone | `bridge/certificates/einstein_maxwell_weyl_opposite_momentum_ell2_tuned_all_primary_bounded_cone.json` |
| All-$\ell$ standard-branch resonance and bounded cone | `bridge/certificates/einstein_maxwell_weyl_symbolic_ell_tuned_axisymmetric_bounded_cone.json` |
| Two-momentum axisymmetric $L = 4$ matrices | `bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_axial_axial_L4_matrix.json` and the three parity companion matrices |
| Two-momentum nonaxisymmetric $L = 3$ matrix | `bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_nonaxisymmetric_L3_matrix.json` |

Continued on next page

| Claim | Authoritative certificate |
|---|---|
| Two-momentum nonaxisymmetric $L = 1, 3$ completion | `bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_nonaxisymmetric_L1_L3_matrix.json` |
| Moment-map/Taub bridge | `bridge/certificates/einstein_maxwell_weyl_moment_map_taub_bridge.json` |

Each certificate records its schema, imported hashes, exact commands, test tier, assumptions, and missing-object ledger. The independent verifier for each named certificate must pass before this manuscript advances to THEOREM-FROZEN. The current draft promotes only the scoped complete formulas displayed above. The candidate-13 real bounded geometry and all fifteen opposite-sign collision carriers are closed. Each same-sign collision cone has a certified nonzero bounded point, while its full real component geometry remains open; so does the unrestricted finite-support bounded zero set.

# 16 Outlook

The immediate mathematical target is the full real component geometry of the six same-sign collision cones. Their nonemptiness, same-fibre shell census, universal four-ray scalar projection, all 24 extreme-ray lifts, and a bounded section over every scalar-cone point are now closed. The complete complex resonance fibre is also stratified over every scalar-cone face. The remaining same-sign gate has now split. On the five nonzero automatic faces, every fixed-occupation rotation-zero link is nonempty and connected. On all six candidates, however, the explicit axisymmetric section lies in the rotation critical locus with Jacobian rank two. The active resonance strata still require a restricted-current nondegeneracy test before any connected-fibre theorem can be applied, while the critical section requires the quadratic normal form of the missing rotation equation. On the automatic faces that normal form is now exact on the complete axial/polar fixed-node-norm tangent. For $N$ occupied nodes of total complex internal dimension $D$, its unquotiented inertia is $(4D-2, 4D-2, 2D-N+2)$, and after node-phase reduction it is $(4D-2, 4D-2, 2D-2N+2)$. Every axisymmetric point also lies on an explicit nonaxisymmetric fixed-occupation arc. The transverse part is therefore hyperbolic on every automatic-face support stratum; what remains locally is to resolve the displayed radical into nonlinear semialgebraic components. On the active candidate-(16) component the separate restricted-current gate is now closed: both resonant nodes are $q_-$, so the complete axial/polar restriction is negative Kähler on every complex smooth stratum, with generic real rank 20. Its singular projective strata and rotation-zero topology remain open. On the candidate-(17)/(20) axisymmetric sections, exact occupation inequalities make the complete Zariski-tangent current nondegenerate, with affine inertia $(6, 10, 0)$ and projective inertia $(5, 9, 0)$; those section points remain algebraically singular. Away from that section, each active variety also contains an exact smooth bounded point with a surviving projective current radical. Hence neither is a global symplectic orbifold; its degeneracy divisor and presymplectic quotient remain open. Candidate (18) has the same global verdict by a different exact mechanism: its two rank-one parity channels have current eigenlines $(1, 1)$ and $(1, -1)$, the scalar cone crosses equal active occupation on the positive mixture $R_3 + s_{18}R_1$, and each resulting smooth bounded family has a four-complex-dimensional projective radical. Thus the common next gate for candidates (17), (18), and (20) is their presymplectic degeneracy divisor and quotient, not another existence witness. The four candidate-(19) and two candidate-(21) real linear sheets are closed separately: every core has current inertia $(5, 5, 0)$ and one connected fixed-occupation rotation-zero link, with spectator factors included. No distinct sheets are identified. The exact fibre-product equation is already certified; complex resonance components and occupation-level sums need not be recomputed. The symbolic-$\ell$ standard-branch resonance and collision theorem is now closed, as is the complete 164-coefficient two-momentum branch-basis census. The next exact enlargement is to classify the real connected components of the fixed-norm resonance strata after imposing all three lifted rotation moment maps. Occupation-level pairwise sums and complex resonance ideals need not be retested. This is followed by joins of distinct $|k|$ fibres without identifying their phase carriers or circumference backgrounds. The immediate physical target is to compare its strata with the transferred Einstein/extra/Maxwell interaction tensor and with observer response. A later causal theorem would replace the finite secular right inverses by support-controlled Green operators or prove that this cannot be done.

The intended final synthesis is not that every fourth-order mode disappears. It is that each mode receives an exact disposition:

$$\boxed{\text{linear mode} \;\to\; \text{moment map} \;\to\; \text{resonance} \;\to\; \text{interaction and observation.}}$$

# References


[1] D. R. Brill and S. Deser, *Instability of closed spaces in general relativity*, Commun. Math. Phys. **32** (1973) 291–304.

[2] A. E. Fischer and J. E. Marsden, *Linearization stability of the Einstein equations*, Bull. Am. Math. Soc. **79** (1973) 997–1003.

[3] V. Moncrief, *Spacetime symmetries and linearization stability of the Einstein equations. I*, J. Math. Phys. **16** (1975) 493–498.

[4] V. Moncrief, *Spacetime symmetries and linearization stability of the Einstein equations. II*, J. Math. Phys. **17** (1976) 1893–1902.

[5] J. M. Arms, *Linearization stability of the Einstein–Maxwell system*, J. Math. Phys. **18** (1977) 830–833.

[6] J. M. Arms, J. E. Marsden, and V. Moncrief, *Symmetry and bifurcations of momentum mappings*, Commun. Math. Phys. **78** (1981) 455–478.

[7] A. E. Fischer, J. E. Marsden, and V. Moncrief, *The structure of the space of solutions of Einstein's equations. I. One Killing field*, Ann. Inst. H. Poincaré A **33** (1980) 147–194.

[8] J. M. Arms, J. E. Marsden, and V. Moncrief, *The structure of the space of solutions of Einstein's equations. II. Several Killing fields and the Einstein–Yang–Mills equations*, Ann. Phys. **144** (1982) 81–106.

[9] E. Lerman, E. Meinrenken, S. Tolman, and C. Woodward, *Non-abelian convexity by symplectic cuts*, Topology **37** (1998) 245–259, arXiv:dg-ga/9603015.

[10] E. Altaş and B. Tekin, *Linearization instability for generic gravity in AdS*, Phys. Rev. D **97** (2018) 024028, arXiv:1705.10234.

[11] E. Altaş and B. Tekin, *Second order perturbation theory in general relativity: Taub charges as integral constraints*, Phys. Rev. D **99** (2019) 104078, arXiv:1903.11982.

[12] A. Pound, *Second-order perturbation theory: problems on large scales*, Phys. Rev. D **92** (2015) 104047, arXiv:1510.05172.

[13] A. Carlotto, *The general relativistic constraint equations*, Living Rev. Relativ. **24** (2021) 2.

[14] J. F. Plebański and S. Hacyan, *Some exceptional electrovac type D metrics with cosmological constant*, J. Math. Phys. **20** (1979) 1004–1010.

[15] J. Lee and R. M. Wald, *Local symmetries and constraints*, J. Math. Phys. **31** (1990) 725–743.

[16] GPT-5.6.sol and Claude Fable 5, *Einstein–Maxwell waves inside Weyl–Maxwell gravity on a compact product*, companion manuscript, 2026.

[17] GPT-5.6.sol, *Pure-extra Taub obstruction and a balanced second-order extension in compact Weyl–Maxwell gravity*, companion manuscript, 2026.